

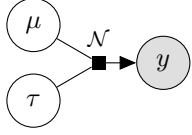
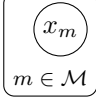
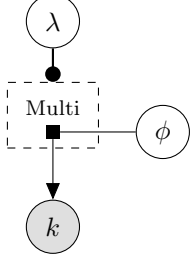
Graphical Models in Tikz

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TikZ examples for graphical models (Bayesian networks) and directed factor graphs [1].

Table 1: Node types

Type	Syntax	Output
Latent variable	<code>\node[latent]</code>	
Observed variable	<code>\node[obs]</code>	
Deterministic	<code>\node[det]</code>	
Constant	<code>\node[const]</code>	a
Factor	<code>\node[factor]</code>	$\mathcal{N}$ 
Factor with nodes		
Plate	<code>\plate</code>	
Gate		

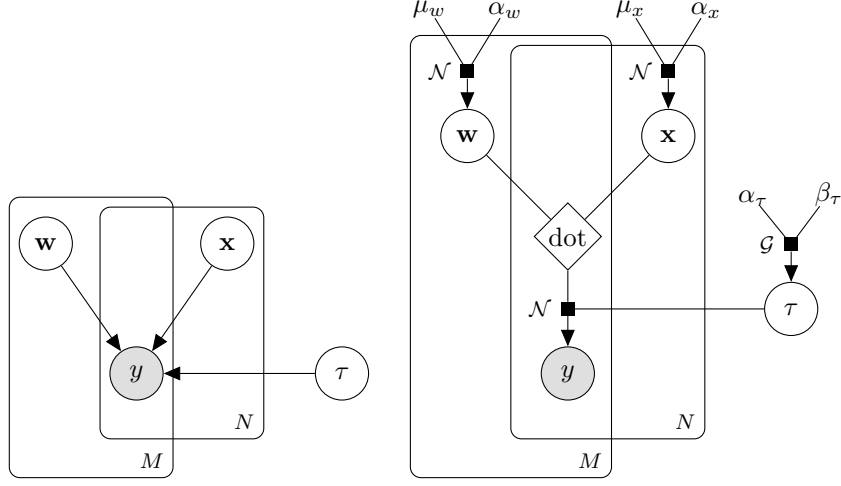


Figure 1: PCA model as a Bayesian network and a directed factor graph.

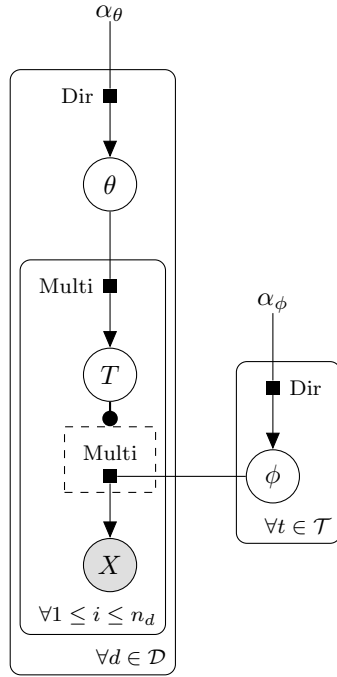


Figure 2: Latent Dirichlet allocation as directed factor graph.

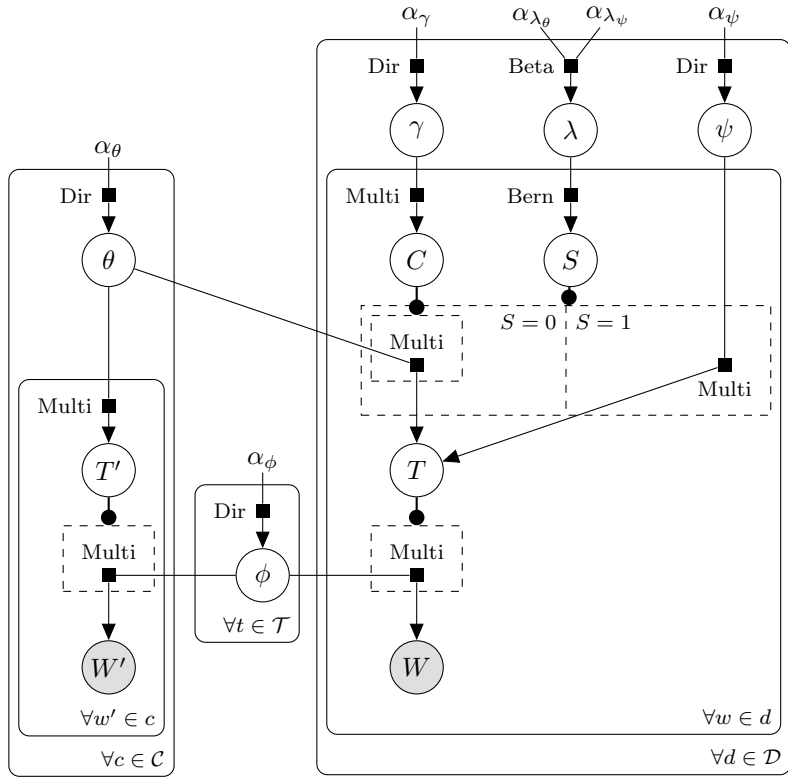


Figure 3: Citation influence model with own topics [2] as directed factor graph.

References

- [1] Laura Dietz, *Directed Factor Graph Notation for Generative Models*. Technical Report. 2010
- [2] Laura Dietz, Steffen Bickel, Tobias Scheffer, *Unsupervised Prediction of Citation Influences*. In: Proceedings of International Conference on Machine Learning. 2007